

STUDENT ACTIVITY FILE – INNOVATIVE DESIGN LAB (IDL) - I

WEEK – 2

A. Personal Interest & Cyber Defense Motivation:

- Deep Interest in System Architecture & Defensive Security. As Team Lead (Gokula Kannan M), my primary focus lies in solving critical engineering bottlenecks where human manual triage speed fails to match machine-speed cyber attacks.

B. Problem Formulation - SOC Alert Fatigue & Telemetry Privacy Dilemma:

- The Alert Triage Bottleneck: Modern Security Operations Centers (SOCs) ingest over 5,000 daily security logs from SIEM agents (Wazuh, Elastic). Tier-1 human analysts spend 30 to 45 minutes investigating single alerts, causing ~70% of security notifications to remain completely unexamined.

- The Cloud Data Leakage Threat: Transmitting un-scrubbed raw telemetry-containing internal IP topographies (RFC 1918), officer emails, hostnames, and credentials-to commercial cloud AI models creates catastrophic data leakage risks, violating privacy mandates such as India's DPDP Act 2023.

C. Project SENTINEL Application & Value Proposition:

- Autonomous Zero-Trust AI Co-Pilot: Delivering machine-speed incident triage (<30s MTTR) via an in-RAM regex data sanitizer, 3-tier model router, and AST code sandbox, ensuring zero cloud telemetry leakage and \$0 Tier-1 compute cost.

A. Technical Skills & Core Competencies (Gokula Kannan M - Team Lead):

- Systems Architecture & Venture Strategy: Expertise in high-throughput asynchronous backend engineering (FastAPI, Python), multi-tier AI model cascading (Ollama local GPU to cloud fallback), and technical project governance.
- Defensive Cybersecurity & Threat Intelligence: Applied knowledge in SIEM rule correlation (Wazuh), MITRE ATT&CK TTP taxonomy mapping, and active defense containment workflows.

B. Critical Societal & Enterprise Needs:

- Enterprise & Healthcare Infrastructure Defense: Small-to-medium enterprise (SME) SOCs, public hospitals, and municipal utilities cannot afford \$100k+/year enterprise SOAR tools (Splunk, Cortex XSOAR), leaving critical infrastructure vulnerable to ransomware.
- Legal Evidence Integrity & Chain-of-Custody: Law enforcement cyber crime cells and air-gapped defense labs require local, tamper-evident JSONL audit logging and courtroom-admissible PDF executive briefs.

C. Bridging the Societal Gap through Project SENTINEL:

- Democratizing Enterprise Security: Providing an open-source, privacy-first AI co-pilot that enables resource-constrained organizations to automate routine threat triage with zero operational cloud licensing overhead.

- Prepare a Passion CV to identify potential entrepreneurial opportunities.

PASSION CV & ENTREPRENEURIAL PROFILE - GOKULA KANNAN M (TEAM LEAD):

- Role & Designation: Systems Architect & Project Lead | Student ID: SEC25CS196 | Department of Computer Science & Engineering (Section A)
- Technical Domain & Core Specialization: Distributed Systems Engineering, AI Pipeline Architecture, Cybersecurity Policy Enforcement, Venture Strategy.
- Key Accomplishments & Leadership Contributions:
 1. System Architecture: Lead architect of Project SENTINEL's 3-Tier AI Router and Zero-Trust Data Sanitizer pipeline.
 2. Venture Pitching & National Validation: Selected to represent Sairam Institutions at Hac'KP 2026 (7th Edition National Hackathon by Kerala Police Cyberdome @ Zoho Corporation Campus).
 3. Research & Academic Output: Co-author of academic literature survey analyzing 6 peer-reviewed IEEE papers on threat triage and privacy-preserving AI.
- Entrepreneurial Opportunity & Commercialization Roadmap:
 - Opportunity Identified: Commercializing SENTINEL as a hybrid open-core / MSSP platform tailored for air-gapped defense labs, municipal SOCs, and police cyber forensic units requiring 100% data sovereignty.